

Morgan Mathison

(816) 787-3567 | morgangmathison@gmail.com

[Personal Research Website](#) | [LinkedIn](#) | <https://orcid.org/0009-0003-7452-2637>

Education

B.S. In Plant Science: Breeding, Biology, and Biotechnology

Summa Cum Laude

University of Missouri - Columbia

May 2025

Minor: Biology | Certification: Retail Floral Management

GPA: 3.96

Research Experience

Postbaccalaureate NSF-RaMP Researcher

University of Kansas - Department of Evolution and Ecology

July 2025- Present

Laboratory of Dr. Brian Atkinson, Co Advised by Dr. Kelly Matsunaga

- Implementing cellulose acetate peeling methodology in order to prepare and section research fossil specimens for analysis
- Utilizing brightfield microscopy to score and describe fossil Cornales fruit for integration into a morphological matrix

Undergraduate Research Assistant

University of Missouri - Division of Plant Science and Technology

May 2023-May 2025

Laboratory of Dr. Jacob Washburn

Thesis: "Utilizing Drone Imaging in Analysis of Genotype by Environment in Maize"

- Operated an unmanned aircraft system to collect hyperspectral and multispectral images of maize fields for further analysis and indice extraction
- Utilized Pix4D and Quantum Geographic Information System to stitch together over 10,000 individual images per drone flight for orthomosaic creation
- Employed Rstudio and Python to extract vegetative indices from orthomosaics to better understand maize health throughout the growing season

Publications in Progress

(Preprint) Mitchell, M., Sidberry, G., **Mathison, M.**, DeSalvio, A. J., Kick, D., & Washburn, J.

D. (2024). Drone-Based Identification of Flood-Tolerant Maize via Multispectral

Imaging: A Real-World Case Study. *bioRxiv*, 2024.11.26.625493.

<https://doi.org/10.1101/2024.11.26.625493>

Honors and Awards

Spring and Fall 2022, 2023, 2024, Spring 2025

University of Missouri - Dean's List

September 2024

Missouri Seedmen's Association Honorary Member Scholarship (**\$1000**)

June 2024

3rd Place Best Poster - University of Missouri Digital Agriculture Symposium

Fall 2021

Southeast Missouri State University - Dean's List

Posters and Presentations

Oral Presentations

Mathison, M. (April, 2025). *Utilizing Drone Imaging to Monitor Crop Health in Maize*. University of Missouri Plant Science Senior Capstone Presentations. Columbia, MO.

Poster Presentations

Mathison, M. (August, 2025.) *Utilizing Drone Imaging in Analysis of Genotype by Environment in Maize*. Kansas Innovation Fest, Overland Park, KS.

Mathison, M. (April, 2025.) *Utilizing Drone Imaging in Analysis of Genotype by Environment in Maize*. University of Missouri Show Me Research Week. Columbia, MO.

Mathison, M. (March, 2025.) *Utilizing Drone Imaging in Analysis of Genotype by Environment in Maize*. Maize Genetics Meeting 2025. St Louis, MO.

Mathison, M. (December, 2024.) *Utilizing Drone Imaging to Understand Plant Health*. University of Missouri Fall Undergraduate Research Day. Columbia, MO.

Mitchell, M., **Mathison, M.** (June, 2024.) *Drone Based Identification of Flood-Tolerant Maize Genotypes*. University of Missouri Digital Agriculture Symposium. Columbia, MO.

Mitchell, M., **Mathison, M.** (April, 2024.) *Drone Based Identification of Flood-Tolerant Maize Genotypes*. University of Missouri Show Me Research Week. Columbia, MO.

Mitchell, M., **Mathison, M.** (March, 2024.) *Drone Based Identification of Flood-Tolerant Maize Genotypes*. University of Missouri Plant Research Symposium. Columbia, MO.

Leadership Experience

Postbaccalaureate NSF-RaMP Researcher

Peer Education Advancement and Research in the Life Sciences (PEARLS)

Co-Creator July 2025- Present

- Co-created PEARLS, a peer cooperative supporting professional growth of postbaccalaureate researchers in the life sciences
- Co-organized "Graduate Experience Panel", facilitating discussion between current graduate students and aspiring scholars in the life sciences

University of Missouri - Columbia

Transfer Experience and Advising Mentors

Director of Leadership Development January - December 2024

- Wrote and sent weekly newsletters to over 400+ club members to inform students of opportunities happening on campus and ways to get involved
- Interacted with stakeholders during weekly meetings to understand club performance and identify ways to better support transfer students

Director of Communications

January - December 2023

- Created content for club and the University of Missouri Transfer Student Center related meetings and activities utilizing Canva for promotion of weekly meetings to increase member participation
- Planned and executed weekly club meetings alongside fellow executive members, providing a weekly space for transfer students to connect with one another

University of Missouri – Columbia

Center for Student Involvement

Welcome Week Leader

August 2023, August 2024

- Supervised over 125+ incoming transfer students during activities the week leading up to the start of fall classes to familiarize students to campus
- Coordinated activity involvement and facilitated participation to introduce new students to one another and build a strong system of connections within the transfer student community

Certifications

Remote Pilot Certification - Small Unmanned Aircraft System

Federal Aviation Administration

Issued February 2024

Certification – Retail Floral Management

University of Missouri – Columbia

May 2025

References

Dr. Brian Atkinson

Postbaccalaureate Research Advisor
Ecology and Evolutionary Biology
University of Kansas
Lawrence, KS, 66045
Brian.Atkinson@ku.edu

Dr. Kelly K.S. Matsunaga

Postbaccalaureate Research Advisor
Ecology and Evolutionary Biology
University of Kansas
Lawrence, KS, 66045
Matsunaga@ku.edu

Dr. Jacob Washburn

Undergraduate Research Advisor
USDA ARS Plant Genetics Research Unit
University of Missouri
Columbia, MO, 65211
Jacob.Washburn@usda.gov

